








Blockchain Applications

Lecture 3: Volatility and Tail Risk



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Today's Question and Route

- **Question**: how much can a crypto position lose tomorrow, and how do we know the risk model is right?
  - an exchange sets margins, a fund reports risk, a firm with crypto on its balance sheet needs a loss budget
- **Route**
  - from volatility clustering (Lecture 2) to volatility models: exponentially weighted moving average (EWMA), generalised autoregressive conditional heteroskedasticity (GARCH)
  - estimation by maximum likelihood with Student-t errors; diagnostics
  - Value-at-Risk (VaR) and Expected Shortfall (ES): four methods
  - backtesting: Kupiec, Christoffersen, the Basel traffic light, an ES check
  - out of sample: the forecasts meet new data
- Data: daily log returns in %, 1 January 2018 – 18 September 2026 (EODHD)

Learning Outcomes

- Write down EWMA and GARCH(1,1) and interpret ω, α, β, persistence and half-life
- Read a GARCH estimation table with Student-t errors and judge the model with residual diagnostics
- Define VaR and ES, and compare Normal, historical, EWMA and GARCH-t estimates
- Interpret the Kupiec, Christoffersen and conditional-coverage tests and the Basel traffic light
- Judge what a short out-of-sample period can and cannot show

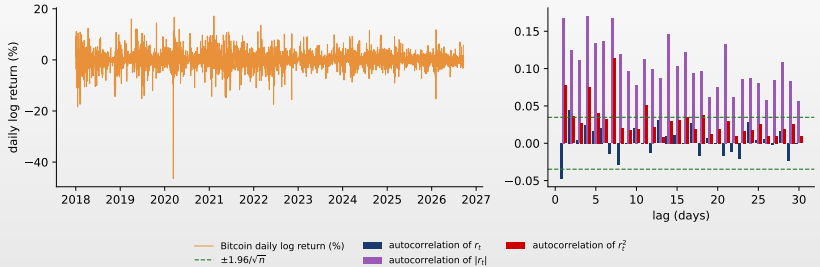
Abbreviations

AIC Akaike information criterion
ATS Advanced Time Series Analysis and Forecasting (course)
BTC Bitcoin
CDF cumulative distribution function
DM Diebold–Mariano (test)
EODHD EOD Historical Data (data vendor)
ES expected shortfall
ETF exchange-traded fund
EWMA exponentially weighted moving average
FABIZ Faculty of Business Administration in Foreign Languages, ASE
GARCH generalised autoregressive conditional heteroskedasticity
GARCH-t GARCH with Student-t errors
GJR Glosten–Jagannathan–Runkle (asymmetric GARCH)
HAC heteroskedasticity- and autocorrelation-consistent
HS historical simulation
IGARCH integrated GARCH
i.i.d. independent and identically distributed
KS Kolmogorov–Smirnov (test)
LR likelihood ratio
MAR market risk standard of the Basel Framework
MSc Master of Science
MV minimum variance
PIT probability integral transform
QLIKE quasi-likelihood loss
RC risk contribution
RMS root mean square
S&P 500 Standard & Poor's 500 stock index
sd standard deviation
se standard error
TLT iShares 20+ Year Treasury Bond ETF
US United States
USD US dollar
VaR value-at-risk

Reading and Tools

- Volatility models: Engle (1982), Bollerslev (1986), Glosten, Jagannathan & Runkle (1993); for crypto: Katsiampa (2017), Chu et al. (2017)
- Risk measures and backtests: Artzner et al. (1999), Kupiec (1995), Christoffersen (1998), McNeil & Frey (2000), Gneiting (2011)
- Regulation: Basel Committee (2019), market-risk standard; Basel Committee, MAR33 (ES), Basel Committee, MAR32 (backtesting)
- Forecast evaluation: Patton (2011), Diebold & Mariano (1995), Berkowitz (2001)
- Video unit on Quantinar: Robust Markowitz for Crypto-Assets
- Related lectures: Advanced Time Series Analysis and Forecasting (ATS), *GARCH Models* and *VaR & ES* (2025); MSc dissertation A.-T. Ginavar, *Risk models for digital assets* (VaR/ES backtests for six cryptos)
- Estimation: Python, package `arch`

Volatility Clustering, Again



- Left: Bitcoin daily log returns, 2018 – 18 September 2026; calm and turbulent periods alternate
- Right: autocorrelations at lags 1–30 days; dashed: $\pm 1.96/\sqrt{n}$, the 5% band under independence
- Returns r_t: 2 of 30 lags outside the band; absolute returns $|r_t|$: all 30 above it (0.06 to 0.17)
- Squared returns r_t^2: 8 of 30 lags above the band; a few extreme days (12 March 2020: -46%) dominate them

From Clustering to a Model

- The chart (and Lecture 2): returns are hardly autocorrelated, absolute and squared returns are autocorrelated
- Model the return as a mean plus a shock with time-varying scale:

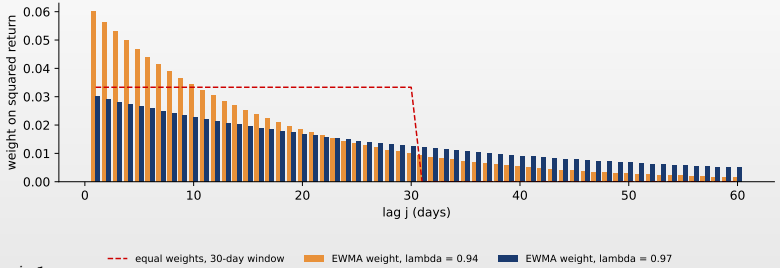
$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \overset{\text{i.i.d.}}{\sim} (0,1)$$

  - $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$: the **conditional variance**, known at the end of day $t-1$
  - $\mathcal{F}_{t-1}$: the information up to day $t-1$
- **EWMA** (RiskMetrics): an exponentially weighted average of past squared shocks

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,\varepsilon_{t-1}^2 = (1-\lambda)\sum_{j\geq 1}\lambda^{j-1}\varepsilon_{t-j}^2, \qquad \lambda = 0.94$$

  - weights decay geometrically; effective memory $1/(1-\lambda) \approx 17$ days
  - no long-run level: after a shock, volatility never returns to a mean

How EWMA Weights the Past



- Weight $(1-\lambda)\lambda^{j-1}$ on the squared return j days ago; red: a 30-day equal-weight window
- $\lambda = 0.94$: 6% on yesterday, 1.0% on day 30 (0.06×0.94^{29}); the first 30 days carry $1 - 0.94^{30} = 84\%$ of the weight
- No cliff at the end of the window: a crash 31 days ago does not suddenly drop out, as it does with a rolling standard deviation

GARCH(1,1)

- Bollerslev (1986):

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0$$

  - α: reaction to yesterday's news; β: memory of yesterday's variance; $\alpha + \beta$: **persistence**
- If $\alpha + \beta < 1$: covariance-stationary, with long-run (unconditional) variance

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}, \qquad \mathrm{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}\left(\sigma_{t+1}^2 - \bar{\sigma}^2\right)$$

- **Half-life** of a volatility shock: $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ days
- $\alpha + \beta = 1$: **integrated** GARCH (IGARCH); shocks never die out; no finite long-run variance
  - with $\omega > 0$: $\mathrm{E}_t[\sigma_{t+h}^2] = \sigma_{t+1}^2 + (h-1)\,\omega$, the forecast rises linearly with h
  - EWMA is the case $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: a flat forecast, $\mathrm{E}_t[\sigma_{t+h}^2] = \sigma_{t+1}^2$

Student-t Errors and Maximum Likelihood

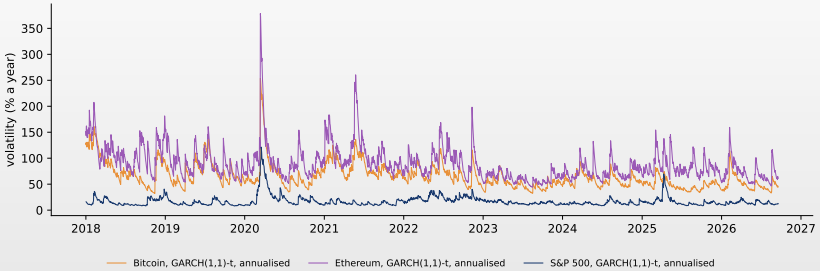
- Heavy tails remain after GARCH filtering: let z_t follow a Student-t with $\nu > 2$ degrees of freedom, scaled to variance 1 (**GARCH-t**)
- Parameters $\theta = (\mu, \omega, \alpha, \beta, \nu)$ by maximum likelihood
  - σ_t^2 computed recursively from θ; $\hat{\theta}$ maximises the Student-t log-likelihood (formula in the Appendix)
  - standard errors: robust (sandwich) covariance of $\hat{\theta}$
- ν measures the tail thickness
  - $\nu \to \infty$: the Normal; ν close to 2: very heavy tails
  - moments of order k exist only for $k < \nu$: with $\nu \leq 4$, $\mathrm{E}[z_t^4]$ is infinite, which matters for the diagnostics

GARCH(1,1)-t Estimates, 2018–2026

	n	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	half-life (days)	$\hat{\nu}$
Bitcoin	3 183	0.144 (0.058)	0.076 (0.012)	0.924 (0.012)	1^*	∞	3.13 (0.15)
Ethereum	3 183	0.550 (0.373)	0.117 (0.038)	0.883 (0.042)	1^*	∞	3.19 (0.20)
S&P 500	2 190	0.035 (0.009)	0.162 (0.022)	0.819 (0.021)	0.981	36	5.85 (0.71)
Gold	2 269	0.015 (0.008)	0.061 (0.018)	0.931 (0.021)	0.992	90	4.06 (0.38)

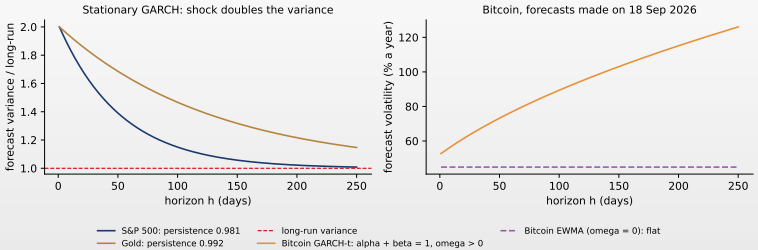
- Daily log returns in %; constant mean; robust standard errors in brackets; * at the bound $\alpha+\beta \le 1$ imposed in estimation
- Interpretation
  - crypto: $\hat{\alpha}+\hat{\beta}=1$ with $\hat{\omega}>0$, an IGARCH; no finite long-run variance, no mean reversion of volatility in this sample
  - S&P 500: a shock to volatility halves in 36 trading days; long-run volatility $\sqrt{252\,\hat{\omega}/(1-\hat{\alpha}-\hat{\beta})}=21.6\%$
  - $\hat{\nu}\approx 3.1$ for crypto, 5.9 for the S&P 500: crypto shocks are heavy-tailed even after filtering

Conditional Volatility



- $\hat{\sigma}_t\sqrt{A}$ from GARCH(1,1)-t, $A = 365$ for crypto, 252 for the S&P 500
- Peaks: Bitcoin 253.5% (13 March 2020), Ethereum 378.4%, S&P 500 121.4% (17 March 2020)
- 18 September 2026: Bitcoin 44.6%, Ethereum 61.2%, S&P 500 12.7%; Bitcoin median 59.2%

After a Shock: Volatility Forecasts by Horizon



- Left, $\hat{\alpha} + \hat{\beta} < 1$: forecast variance relative to the long-run level, after a shock that doubles the variance
  - S&P 500 (0.981): half-life $\ln 0.5 / \ln 0.981 = 36.1$ days; after 250 days the shock is gone
  - gold (0.992): half-life 90 days; still 15% above the long run after 250 days
- Right, Bitcoin, forecasts made on 18 September 2026: no long-run level to return to
  - GARCH-t ($\hat{\alpha} + \hat{\beta} = 1$, $\hat{\omega} = 0.144$): the variance grows by $\hat{\omega}$ a day; volatility 52.7% a year at $h = 1$, 126% at $h = 250$
  - EWMA ($\omega = 0$): flat at 44.9%; only here does $\sqrt{h}$ scaling of today's volatility hold exactly

Diagnostics: Did the Model Remove the Clustering?

- Standardised residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$ should be i.i.d.: Ljung–Box $Q(10)$ of $\hat{z}_t^2$ against χ^2_{10} (5% critical value 18.31)

	$Q(10)$ of r_t^2	$Q(10)$ of $\hat{z}_t^2$	p-value for $\hat{z}_t^2$ (χ^2_{10})
Bitcoin	97.6	5.5	0.855
Ethereum	131.8	8.8	0.552
S&P 500	2 338.8	12.0	0.283
Gold	434.4	27.9	0.002

- Interpretation
  - crypto and the S&P 500: we do not detect autocorrelation in $\hat{z}_t^2$ at lags 1–10; this does not prove that all dependence is gone
  - gold: some dependence remains (p 0.002); a richer model (asymmetry, regime changes in 2025–2026) is needed
  - caveat: the χ^2_{10} reference needs $\mathrm{E}[z_t^4] < \infty$; with $\hat{\nu} \approx 3$ it is infinite (next slide)
  - sample 2018–2026; the $Q(10) = 212.7$ of Lecture 2 used the longer sample since 2014

Diagnostics with Heavy Tails

- Two checks that do not need $\mathrm{E}[z_t^4] < \infty$
  - permutation p-value: reshuffle $\hat{z}_t$ 999 times; share of reshuffled $Q(10)$ of $\hat{z}_t^2$ at least as large as the observed one
  - Ljung–Box $Q(10)$ of $|\hat{z}_t|$: needs only a finite variance of $|z_t|$, i.e. $\nu > 2$

	$Q(10)$ of $\hat{z}_t^2$	permutation p	$Q(10)$ of $\lvert\hat{z}_t\rvert$	p-value (χ^2_{10})
Bitcoin	5.5	0.232	24.5	0.006
Ethereum	8.8	0.291	10.5	0.395
S&P 500	12.0	0.244	8.3	0.597
Gold	27.9	0.012	41.8	< 0.001

- Interpretation
  - the χ^2 p-value for Bitcoin's $\hat{z}_t^2$ (0.855) is far from the permutation p-value (0.232): the usual reference is unreliable here
  - Bitcoin: $|\hat{z}_t|$ is still autocorrelated (p 0.006); GARCH(1,1)-t leaves some clustering in the size of the shocks
  - Ethereum and the S&P 500: no remaining dependence detected; gold: rejected by all three checks
  - all p-values treat $\hat{\theta}$ as known; estimation error is ignored

VaR and ES: Definitions

- For a one-day return r_{t+1} with conditional distribution F_t and level p (e.g. 1%):

$$\mathrm{VaR}_p = -F_t^{-1}(p), \qquad \mathrm{ES}_p = -\mathrm{E}\big[r_{t+1} \mid r_{t+1} \le -\mathrm{VaR}_p\big] = \frac{1}{p}\int_0^p \mathrm{VaR}_u \, du$$

  - VaR: the loss exceeded with probability p; ES: the average loss when VaR is exceeded
- Artzner et al. (1999): a **coherent** risk measure is monotone, translation-invariant, positively homogeneous and **subadditive**: $\rho(X+Y) \le \rho(X) + \rho(Y)$
  - ES is coherent; VaR is not subadditive in general, so diversification can appear to raise VaR
  - ES sees the size of the tail; VaR only its threshold
- Regulation (Basel Committee (2019)): for banks using internal models for market-risk capital, ES 2.5% replaced VaR 1% (Basel Committee, MAR33)
  - VaR 1% stays in the backtesting of these models (Basel Committee, MAR32)
  - under the Normal: $\mathrm{VaR}_{1\%} = 2.326\,\sigma$ and $\mathrm{ES}_{2.5\%} = 2.338\,\sigma$, almost equal; under heavy tails ES is larger

Four Ways to Compute VaR and ES

- **Normal**: $\mathrm{VaR}_p = -(\hat{\mu} + \hat{\sigma}\,\Phi^{-1}(p))$, $\mathrm{ES}_p = -\hat{\mu} + \hat{\sigma}\,\varphi(\Phi^{-1}(p))/p$
  - Φ, φ: the standard Normal distribution function and density; $\Phi^{-1}(p)$: its p-quantile
- **Historical simulation** (HS): $\mathrm{VaR}_p = -\hat{q}_p$, the empirical p-quantile of the last W returns; ES_p = minus the mean of returns below $\hat{q}_p$
- **EWMA**: Normal formulas with $\hat{\sigma}_{t+1}$ from the EWMA recursion, $\hat{\mu} = 0$
- **GARCH-t**: with $q = t_\nu^{-1}(p)$ and $s = \sqrt{(\nu-2)/\nu}$

$$\mathrm{VaR}_p = -\big(\hat{\mu} + \hat{\sigma}_{t+1}\, s\, q\big), \qquad \mathrm{ES}_p = -\hat{\mu} + \hat{\sigma}_{t+1}\, s\, \frac{f_\nu(q)}{p}\,\frac{\nu + q^2}{\nu - 1}$$

  - t_ν^{-1}, f_ν: quantile function and density of the standard Student-t; s rescales the t to variance 1
- **Horizon**: $\mathrm{VaR}_h \approx \sqrt{h}\,\mathrm{VaR}_1$ only for i.i.d. Normal returns; with GARCH, use the variance forecasts $\mathrm{E}_t[\sigma^2_{t+j}]$

Worked Example: GARCH-t VaR and ES for Bitcoin

- Inputs on 18 September 2026: $\hat{\sigma}_{t+1} = 2.761\%$, $\hat{\mu} = 0.06\%$, $\hat{\nu} = 3.126$
- Scale factor $s = \sqrt{(\nu-2)/\nu} = \sqrt{1.126/3.126} = 0.600$
- VaR 1%: $q = t_{3.126}^{-1}(0.01) = -4.398$

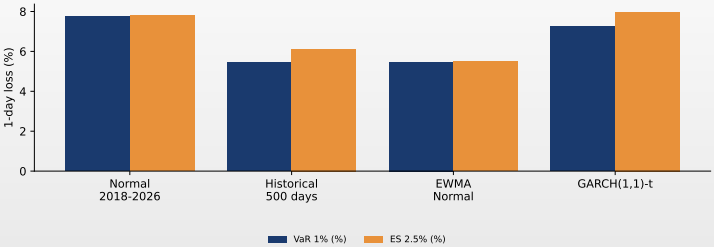
$$\mathrm{VaR}_{1\%} = -0.06 + 2.761 \times 0.600 \times 4.398 = 7.23\%$$

- ES 2.5%: $q = t_{3.126}^{-1}(0.025) = -3.111$; $\dfrac{f_\nu(q)}{p}\dfrac{\nu+q^2}{\nu-1} = 4.844$

$$\mathrm{ES}_{2.5\%} = -0.06 + 2.761 \times 0.600 \times 4.844 = 7.97\%$$

- Under the Normal with the same σ: $2.761 \times 2.338 = 6.45\%$; the t tail adds 1.5 percentage points

VaR and ES by Method



- Bitcoin, one-day horizon, forecasts for 19 September 2026 (table on the next slide)
- Normal and EWMA: ES $\approx$ VaR; historical and GARCH-t: ES exceeds VaR by 13% and 10%

Bitcoin: VaR and ES for 19 September 2026

Method	VaR 1% (%)	ES 2.5% (%)	VaR, USD per 1m	ES, USD per 1m
Normal, 2018–2026	7.75	7.79	74 567	74 862
Historical simulation, 500 days	5.43	6.12	52 854	59 002
EWMA, $\lambda = 0.94$, Normal	5.47	5.49	53 208	53 433
GARCH(1,1)-t	7.23	7.97	69 753	75 761

- One-day losses of a long position of 1 million USD, in log-return % and in USD
  - VaR in USD: $10^{6}(1 - e^{-\mathrm{VaR}/100})$, exact, since the USD loss is a monotone function of the return
  - ES in USD: the mean of the USD loss $10^{6}(1 - e^{r/100})$ over the 2.5% tail of r, not $10^{6}(1 - e^{-\mathrm{ES}/100})$
- Interpretation
  - the Normal and EWMA give $\mathrm{ES}_{2.5\%} \approx \mathrm{VaR}_{1\%}$: they ignore the heavy tail
  - GARCH-t: ES exceeds VaR by 10%, the cost of the heavy tail ($\hat{\nu} = 3.13$)
  - the four VaR figures differ by up to 43%: the choice of model is itself a risk (**model risk**)

Backtesting VaR: the Hit Sequence

- Hit (violation): $I_t = \mathbb{1}\{r_t < -\mathrm{VaR}_{p,t}\}$; a correct model has $I_t \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$
- **Unconditional coverage** Kupiec (1995): x hits in n days, $\hat{\pi} = x/n$

$$\mathrm{LR}_{uc} = -2 \ln \frac{(1-p)^{n-x} p^x}{(1-\hat{\pi})^{n-x} \hat{\pi}^x} \sim \chi^2_1$$

- **Independence** Christoffersen (1998): n_{ij} = number of days in state j after state i; $\hat{\pi}_{ij} = n_{ij}/(n_{i0} + n_{i1})$

$$\mathrm{LR}_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1-\hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi^2_1$$

- **Conditional coverage**: $\mathrm{LR}_{cc} = \mathrm{LR}_{uc} + \mathrm{LR}_{ind} \sim \chi^2_2$
- **Basel traffic light** (Basel Committee, MAR32), 250 days at VaR 1%: green 0–4 hits, amber 5–9 ("yellow" in the 1996 rules), red 10 or more

Worked Example: the Kupiec Test

- Normal VaR on a 250-day window, Bitcoin 2020–2026: $n = 2\,453$ days, $x = 45$ hits, $p = 0.01$, $\hat{\pi} = 45/2\,453 = 0.0183$

$$\ln L_0 = 2\,408 \ln 0.99 + 45 \ln 0.01 = -231.4, \qquad \ln L_1 = 2\,408 \ln(1 - 0.0183) + 45 \ln 0.0183 = -224.5$$

$$\mathrm{LR}_{uc} = -2(\ln L_0 - \ln L_1) = 13.84 > 3.84$$

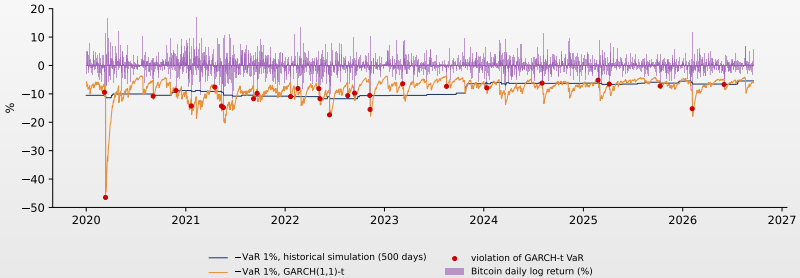
- The model is rejected: 45 hits where 24.5 were expected, almost twice the promised frequency
- For a bank this is a capital issue: Basel multiplies market-risk capital by a factor that rises with the number of exceptions

Backtest: Bitcoin, 1-Day VaR 1%, 2020–2026

Method	hits	rate (%)	LR_{uc} (p)	LR_{ind} (p)	LR_{cc} (p)	last 250	zone	mean VaR (%)
Normal, 250 days	45	1.83	13.84 (<0.001)	0.04 (0.850)	13.88 (0.001)	5	amber	7.16
HS, 500 days	25	1.02	0.01 (0.924)	1.29 (0.256)	1.30 (0.523)	3	green	8.64
EWMA, Normal	51	2.08	22.01 (<0.001)	0.00 (0.952)	22.01 (<0.001)	5	amber	6.78
GARCH(1,1)-t	28	1.14	0.47 (0.491)	0.95 (0.329)	1.43 (0.490)	2	green	8.12

- 2 453 out-of-sample days, 1 January 2020 – 18 September 2026; expected hits $np = 24.5$
- GARCH-t re-estimated every 21 days on all data up to the previous day, then filtered daily
- Interpretation
  - Normal and EWMA fail unconditional coverage: almost twice as many hits as promised
  - HS and GARCH-t: none of the three tests rejects; GARCH-t reacts to news, HS needs a higher average VaR (8.64% vs 8.12%) for similar coverage

Violations Through Time



- The HS line moves in steps and lags the market; the GARCH-t line widens within days after a shock
- Hits of GARCH-t are spread over time: no clustering, in line with LR_{ind} (p 0.329)

Checking Expected Shortfall

- ES is not elicitable on its own (Gneiting (2011)); a simple check in the spirit of McNeil & Frey (2000): on the days beyond the 2.5% quantile, compare realised loss and predicted ES

$$\bar{R} = \frac{1}{N} \sum_{t:\, r_t < -\mathrm{VaR}_{\mathbf{2.5\%},t}} \frac{-r_t}{\mathrm{ES}_{\mathbf{2.5\%},t}}, \qquad \text{correct model: } \bar{R} \approx 1$$

Method	tail days N	rate (%)	mean loss (%)	mean ES (%)	$\bar{R}$	95% interval for $\bar{R}$
Normal, 250 days	71	2.89	8.48	6.72	1.26	1.13 – 1.44
HS, 500 days	61	2.49	9.05	9.14	1.00	0.89 – 1.15
EWMA, Normal	76	3.10	7.94	6.11	1.31	1.17 – 1.49
GARCH(1,1)-t	76	3.10	7.92	8.04	0.97	0.88 – 1.09

- Interval: bootstrap of the N ratios (9 999 resamples), tail days treated as independent; expected tail days $0.025 \times 2\,453 = 61.3$
- Normal-based ES: realised tail losses are 26–31% larger than predicted; both intervals lie above 1
- HS and GARCH-t: the intervals contain 1; no evidence against their ES level in this sample
- GARCH-t: 76 tail days against 61.3 expected (Kupiec at 2.5%: $\mathrm{LR}_{uc} = 3.35$, p 0.067)
- A preliminary diagnostic, not a formal ES backtest

The Forecasts Meet New Data

- All estimates in this lecture use data to 18 September 2026 (day T); EODHD data to 23 October 2026 are now available
- GARCH-t parameters are **frozen** on 18 September; only the filter is updated with new returns

$$\hat{\sigma}^2_{t|t-1} = \hat{\omega} + \hat{\alpha}\,(r_{t-1} - \hat{\mu})^2 + \hat{\beta}\,\hat{\sigma}^2_{t-1|t-2}, \qquad t = T+1, \ldots, T+H$$

- Two questions
  - Was the *level* of volatility right? realised $\sqrt{A\,\overline{r^2}}$ against the 18 September forecast $\sqrt{A\,\frac{1}{H}\sum_h E_T \sigma^2_{T+h}}$
  - Did the VaR hold? hits against *np*
- 35 calendar days for crypto, 25 trading days for the S&P 500 and gold: a short window, so the question is also *how much can 35 days tell us?*

What Is Updated after 18 September

Method	parameters	volatility state	estimation window
GARCH-t	fixed on 18 Sep	updated daily with r_{t-1}	2018 – 18 Sep, fixed
EWMA, $\lambda = 0.94$	λ fixed	updated daily with r_{t-1}	none
Normal, VaR rolled forward	mean and sd re-estimated daily	none	2018 to $t-1$, expanding
Normal, volatility table	fixed on 18 Sep	none	2018 – 18 Sep, fixed
HS, 500 days	none	none	last 500 days to $t-1$, rolling

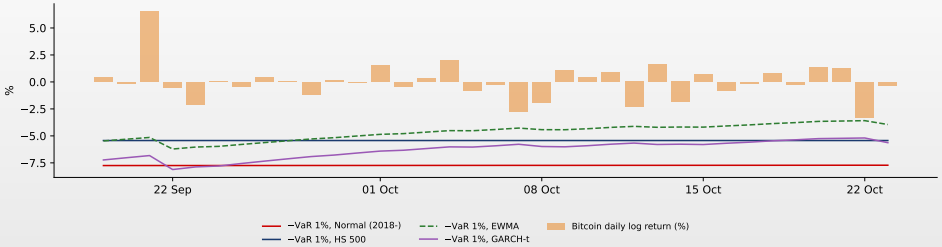
- Volatility level (next slide): forecasts made on 18 September for the whole window, never updated
- One-step forecasts (VaR rolled forward; Appendix: quasi-likelihood (QLIKE) loss, probability integral transform (PIT)): updated as in the table
- The comparison is between forecasts built differently: GARCH-t and EWMA react to new returns, the Normal and HS only through their windows

Volatility: Forecast against Realised

annualised, %	Bitcoin	Ethereum	S&P 500	Gold
return over the window (%)	0.2	−9.1	−0.4	−9.3
GARCH-t $\hat{\sigma}_{T+\mathbf{1}}$, 18 Sep	52.7	72.6	11.9	23.3
GARCH-t, average over the window	60.6	93.2	14.4	23.2
EWMA, 18 Sep	44.9	58.8	10.4	23.9
Normal, 2018–2026	64.1	84.3	19.2	16.9
realised, 19 Sep – 23 Oct	31.8	36.1	10.3	19.3
realised / GARCH forecast	0.52	0.39	0.72	0.83

- Every model over-predicted: Bitcoin's realised 31.8% equals the calm of August 2026
- Bitcoin and Ethereum: $\hat{\alpha} + \hat{\beta} = 1$ with $\hat{\omega} > 0$, so the 18 September forecast *rises* linearly in variance; there is no long-run level to revert to
- Bitcoin: from 52.7% for 19 September to 67.6% on day 35 of the window, 60.6% on average
- Gold fell 9.3% with realised volatility 19.3%, below its GARCH forecast (23.2%): a large price move is not the same as high volatility

Bitcoin VaR, Rolled Forward



1-day VaR 1%, 35 days	Normal	HS 500	EWMA	GARCH-t
mean VaR (%)	7.73	5.43	4.66	6.28
mean capital per 1m USD	74 428	52 854	45 500	60 871
hits (expected 0.35)	0	0	0	0

- Worst day −3.27% (22 October): 0.63 of that day's GARCH-t VaR; 18 September forecast for 19 September: VaR 7.23%, realised +0.41%

How Much Can 35 Days Tell Us?

- With a correct 1% VaR, the number of hits is $X \sim \mathrm{Bin}(35, 0.01)$:

$$P(X = 0) = 0.99^{35} = 0.703, \qquad \mathrm{LR}_{uc}(X = 0) = -2 \cdot 35 \ln 0.99 = 0.70 \; (p = 0.40)$$

- A model with a *true* hit rate of 2% (as Normal-250 in the backtest) still shows zero hits with probability $0.98^{35} = 0.493$
- Power: to detect 2% against 1% with probability 0.8 at the 5% level (exact one-sided binomial test) one needs about **900 days** (905)
- Conclusions
  - Zero hits in five weeks is *no evidence* for any of the four methods: all four "pass"
  - The methods differ in capital: GARCH-t asks for 35% more than EWMA in this window; the long backtest, not this window, says which one is right
  - Use the whole density, not only the hits (Appendix)

Summary

- GARCH(1,1)-t fits crypto with $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH) and $\hat{\nu} \approx 3.1$: shocks are very heavy-tailed even after filtering
- Diagnostics: no autocorrelation detected in Bitcoin's $\hat{z}_t^2$; with $\hat{\nu} \approx 3$ the check on $|\hat{z}_t|$ still finds some (p 0.006)
- Bitcoin 1-day VaR 1% on 19 September 2026: 5.4% to 7.8% depending on the method; ES 2.5% up to 8.0%
- Backtest 2020–2026: Normal and EWMA fail Kupiec (45 and 51 hits for 24.5 expected); for HS and GARCH-t no test rejects
- With Normal-based ES, realised tail losses are 26–31% larger than predicted (95% intervals above 1)
- Out of sample (to 23 October 2026) all models over-predicted volatility; 35 days cannot discriminate between VaR methods

Interpretation Exercises

1. Bitcoin: $\hat{\alpha} + \hat{\beta} = 1$ (at the bound); S&P 500: 0.981. What does each imply for a 10-day volatility forecast made after a crash?
2. Ljung–Box for gold: $Q(10)$ of $\hat{z}_t^2 = 27.9$ (p 0.002). Is the model adequate? What would you try next?
3. A risk manager uses EWMA-Normal for Bitcoin and had 5 violations in the last 250 days. Which zone is that, and what do the full-period tests say about the model?
4. The GARCH-t ES for 1 million USD is 75 761 USD and the Normal ES is 74 862 USD, yet the backtest favours GARCH-t clearly. Explain using $\bar{R}$.
5. Out of sample (19 Sep – 23 Oct 2026) no method had a Bitcoin VaR hit. A manager concludes that the Normal model is now fine. Respond with the binomial argument.

- Next: Lecture 4, stablecoins and depeg risk
  - a depeg is described by its size, its duration and its probability
  - a daily VaR of returns gives only a loss threshold for one day: none of the three

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- The Student-t Log-Likelihood
- Is There a Leverage Effect?
- News Impact Curves
- Diagnostics: Normal or Student-t Errors?
- The Traffic Light Through Time
- Euler Risk Contributions
- Worked Example: Euler Contributions with Two Assets
- A 60/40 Portfolio with Bitcoin
- Risk Share against Weight
- Does Bitcoin Improve the Efficient Frontier?
- One-Step Volatility Forecasts, Day by Day
- Which Forecast Is Best? QLIKE and Diebold–Mariano
- Was the Density Right? PIT and the Berkowitz Test
- The Portfolio: Correlation Is the Unstable Input

The Student-t Log-Likelihood

- Log-likelihood of $\theta = (\mu, \omega, \alpha, \beta, \nu)$, with $\varepsilon_t = r_t - \mu$ and σ_t^2 from the GARCH recursion:

$$\ell(\theta) = \sum_{t=1}^{n} \left[\ln \Gamma\left(\tfrac{\nu+1}{2}\right) - \ln \Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2} \ln\left(\pi(\nu-2)\sigma_t^2\right) - \tfrac{\nu+1}{2} \ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right) \right]$$

  - $\Gamma(x) = \int_0^\infty u^{x-1} e^{-u}\, du$: the gamma function
  - each term is the log density of a Student-t with ν degrees of freedom, scaled to variance σ_t^2
  - $\hat{\theta} = \arg\max \ell(\theta)$ by numerical optimisation, subject to $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta \leq 1$, $\nu > 2$

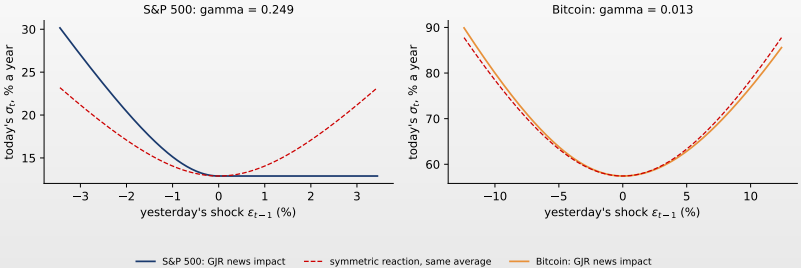
Is There a Leverage Effect?

- GJR-GARCH (Glosten, Jagannathan & Runkle (1993)): $\sigma_t^2 = \omega + \big(\alpha + \gamma\, 1\{\varepsilon_{t-1} < 0\}\big)\varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2$
  - $\gamma > 0$: a fall raises volatility more than a rise of the same size; persistence $\alpha + \beta + \gamma/2$
  - test $\gamma = 0$: $\mathrm{LR} = 2(\ell_{\mathbf{GJR}} - \ell_{\mathbf{GARCH}}) \sim \chi_1^2$, 5% critical value 3.84
  - Akaike information criterion $\mathrm{AIC} = 2k - 2\ell$, k = number of parameters: lower AIC is better fit after a penalty for the extra γ

	$\hat{\gamma}$	$t(\hat{\gamma})$	$\hat{\alpha} + \hat{\beta} + \hat{\gamma}/2$	LR	p-value	AIC: GARCH / GJR
Bitcoin	0.013	0.72	1.000	0.68	0.408	15 518 / 15 520
Ethereum	0.028	0.74	0.998	1.19	0.275	17 493 / 17 494
S&P 500	0.249	6.25	0.968	75.89	< 0.001	5 902 / 5 828
Gold	-0.060	-3.21	0.995	14.27	< 0.001	5 980 / 5 968

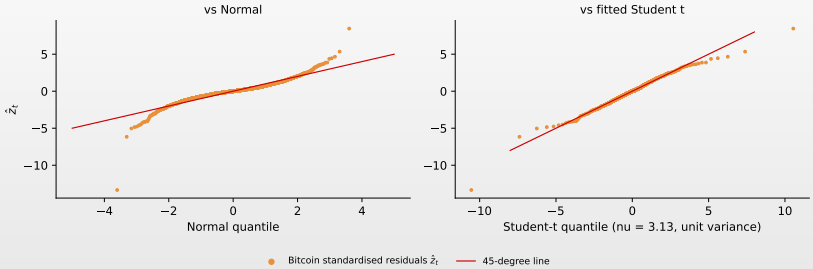
- S&P 500: falls raise volatility much more than rises; GJR preferred by LR and AIC
- gold: $\hat{\gamma} < 0$: rises raise volatility more than falls
- crypto: no asymmetry detected; AIC prefers GARCH(1,1)

News Impact Curves



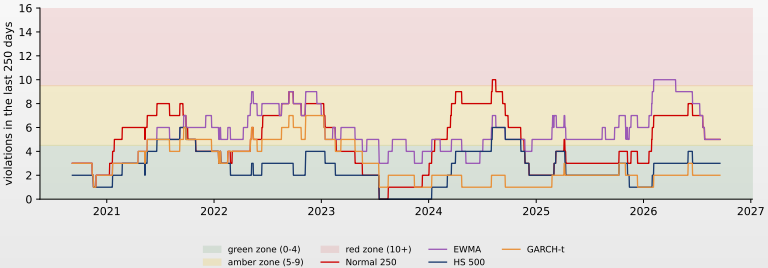
- Today's volatility as a function of yesterday's shock, with yesterday's variance at its sample median (GJR-GARCH-t)
- S&P 500: a -3% day raises volatility to 27.0% a year, a $+3\%$ day leaves it at 12.9%: strong asymmetry
- Bitcoin: the curve is almost symmetric ($\hat{\gamma} = 0.013$): good and bad news raise volatility alike

Diagnostics: Normal or Student-t Errors?



- Bitcoin $\hat{z}_t$ against Normal quantiles (left) and against the fitted Student-t with $\hat{\nu} = 3.13$ (right)
- The Normal fails in both tails; the Student-t fits well except the single most extreme day (12 March 2020: $\hat{z}_t = -13.3$ against a t quantile of -10.5)

The Traffic Light Through Time



- Number of hits in the last 250 days for each method; background: Basel green, amber and red zones
- Share of days in the amber or red zone: Normal 50.3%, EWMA 75.6% (red 3.4%), historical 10.5%, GARCH-t 22.9% (never red)
- A model can pass a full-sample test and still spend long periods in the amber zone: monitor it continuously

Euler Risk Contributions

- Portfolio return $r_p = w^\top r$, weights w, covariance matrix Σ; volatility $\sigma_p = \sqrt{w^\top \Sigma w}$
- σ_p is homogeneous of degree 1 in w, so by Euler's theorem it is the sum of the risk contributions (RC)

$$\sigma_p = \sum_i w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_i \mathrm{RC}_i, \qquad \mathrm{RC}_i = \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- The same holds for ES; with historical data

$$\mathrm{ES}_p = \sum_i \mathrm{RC}_i^{ES}, \qquad \mathrm{RC}_i^{ES} = -\mathrm{E}\big[w_i r_i \mid r_p \le q_p(r_p)\big]$$

- RC_i/σ_p: the share of risk due to asset i; compare it with the weight w_i
- Minimum-variance (MV) portfolio: $w^{\mathrm{MV}} = \Sigma^{-1}1/(1^\top \Sigma^{-1}1)$

Worked Example: Euler Contributions with Two Assets

- 95% S&P 500 ($\sigma_1 = 19.2\%$), 5% Bitcoin ($\sigma_2 = 63.3\%$), correlation $\rho = 0.29$; daily simple returns 2018–2026, annualised

$$\sigma_p^2 = 0.95^2 \cdot 19.2^2 + 0.05^2 \cdot 63.3^2 + 2 \cdot 0.95 \cdot 0.05 \cdot 0.29 \cdot 19.2 \cdot 63.3 = 376.2, \quad \sigma_p = 19.40\%$$

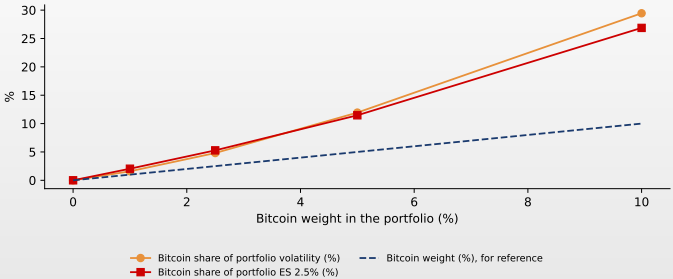
- Contributions $\mathrm{RC}_i = w_i(\Sigma w)_i / \sigma_p$
  - S&P 500: 18.02 percentage points (92.9%); Bitcoin: 1.38 points (7.1%)
  - check: $18.02 + 1.38 = 19.40 = \sigma_p$
- Bitcoin carries 7.1% of the risk with 5% of the money; in the 60/40 portfolio (next slide) it carries 11.9%, because Treasuries diversify the equity risk but not the Bitcoin risk

A 60/40 Portfolio with Bitcoin

Bitcoin weight (%)	volatility (% a year)	ES 2.5% (%, daily)	Bitcoin share of volatility (%)	Bitcoin share of ES (%)
0	12.25	2.26	0.0	0.0
1	12.31	2.29	1.6	2.0
2.5	12.45	2.32	4.8	5.3
5	12.83	2.40	11.9	11.5
10	14.08	2.60	29.4	26.9

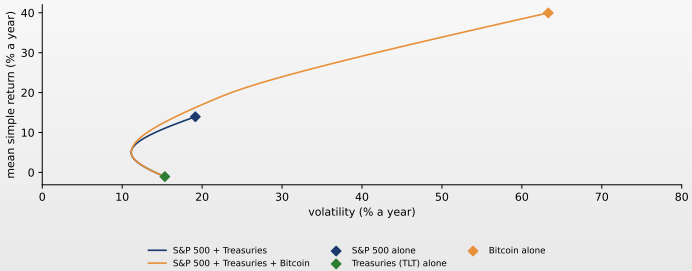
- S&P 500 and long US Treasuries in proportion 60/40, plus Bitcoin; 2 190 common days, 2018–2026
  - S&P 500: index GSPC, a price index, so dividends are not included
  - Treasuries: exchange-traded fund (ETF) TLT, adjusted close, so distributions are reinvested
  - daily simple returns; portfolio return $w^\top R_t$, weights rebalanced daily
- Correlations: Bitcoin–S&P 500 0.29, Bitcoin–Treasuries -0.02, S&P 500–Treasuries -0.14
- A 5% Bitcoin weight carries 12% of the portfolio volatility; 10% carries 29%
- Minimum-variance weights: S&P 500 40.5%, Treasuries 59.6%, Bitcoin -0.15%: Bitcoin does not reduce risk

Risk Share against Weight



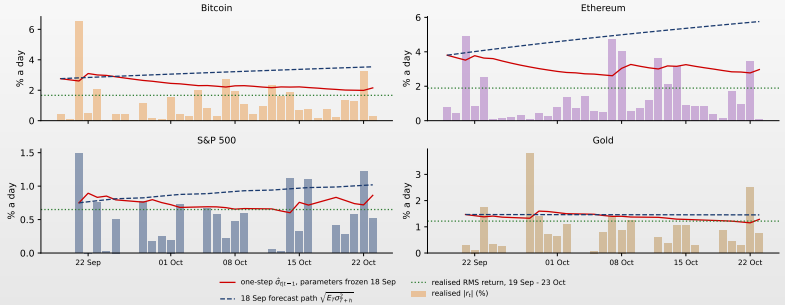
- The risk share grows faster than the weight: from 1.6 times the weight at 1% to 2.9 times at 10%
- A position limit for crypto should be set on its risk contribution, not on its weight
- Robust alternatives to the sample covariance: Robust Markowitz for Crypto-Assets

Does Bitcoin Improve the Efficient Frontier?



- Long-only minimum-variance frontiers, 2018–2026 common days; mean daily simple return $\times 252$ and volatility, both annualised
- Mean return and volatility: Bitcoin 40.0% at 63.3%; S&P 500 13.9% at 19.2% (no dividends); TLT -1.1% at 15.3%
- Compound growth of the net asset value: Bitcoin 21.7%, S&P 500 12.8%, TLT -2.2% a year
- Adding Bitcoin extends the frontier only at high volatility; at the low-risk end the two frontiers coincide: in-sample means, so an upper bound on the benefit

One-Step Volatility Forecasts, Day by Day



- Red: GARCH-t with frozen parameters reacts to each $|r_t|$ (Bitcoin +6.5% on 21 September); blue: the 18 September term structure
- Ethereum's path goes up: $\hat{\alpha} + \hat{\beta} = 1$ (at the bound) and $\hat{\omega} = 0.55 > 0$, so the forecast variance grows by $\hat{\omega}$ each day

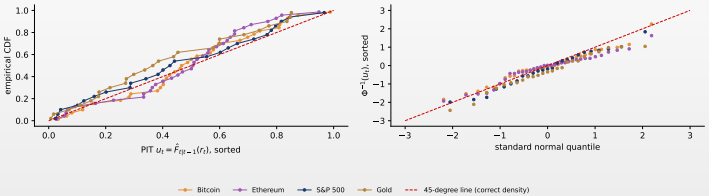
Which Forecast Is Best? QLIKE and Diebold–Mariano

- Quasi-likelihood (QLIKE) loss, robust to the noise in r_t^2 as a proxy of σ_t^2 (Patton (2011)): $L_t(h) = \ln h_t + r_t^2/h_t$
- Diebold & Mariano (1995): $d_t = L_t(\text{GARCH}) - L_t(\text{other})$, $\mathrm{DM} = \bar{d}/\widehat{\mathrm{se}}_{\mathrm{HAC}}(\bar{d}) \sim N(0,1)$, HAC: heteroskedasticity- and autocorrelation-consistent; $\bar{d} > 0$: GARCH loses

	Bitcoin	Ethereum	S&P 500	Gold
mean QLIKE: GARCH-t	2.232	2.655	0.220	1.480
mean QLIKE: EWMA	2.094	2.435	0.184	1.480
mean QLIKE: constant (Normal)	2.667	3.152	0.670	1.431
GARCH vs EWMA: $\bar{d}$ (DM)	0.138 (1.97)	0.220 (2.99)	0.036 (0.83)	−0.000 (−0.01)
GARCH vs constant: $\bar{d}$ (DM)	−0.435 (−4.11)	−0.497 (−6.12)	−0.450 (−3.50)	0.048 (0.32)

- A time-varying forecast beats the constant one clearly for Bitcoin, Ethereum and the S&P 500
- EWMA beats GARCH for Ethereum ($\mathrm{DM} = 2.99$) and, at the margin, for Bitcoin: it forgets faster, and after a calm month that helps
- But in the 2020–2026 backtest EWMA-Normal failed the VaR test: being good at the *level* is not being good at the *tail*

Was the Density Right? PIT and the Berkowitz Test



- PIT $u_t = \hat{F}_{t|t-\mathbf{1}}(r_t)$: i.i.d. $U(0,1)$ if the density is right; Berkowitz (2001): $x_t = \Phi^{-\mathbf{1}}(u_t)$, fit $x_t = c + \rho x_{t-\mathbf{1}} + \sigma\varepsilon_t$, $\mathrm{LR} = 2(\ell_{\mathbf{1}} - \ell_{\mathbf{0}}) \sim \chi^{\mathbf{2}}_{\mathbf{3}}$ under $c = 0, \rho = 0, \sigma = 1$

	Bitcoin	Ethereum	S&P 500	Gold
sd of x_t (should be 1)	0.84	0.78	0.98	0.93
Berkowitz LR (p)	3.16 (0.368)	4.80 (0.187)	2.36 (0.501)	5.59 (0.133)
Kolmogorov–Smirnov D (p)	0.125 (0.604)	0.176 (0.203)	0.119 (0.833)	0.186 (0.310)

- sd of x_t below 1 for crypto: predicted densities too wide (over-predicted volatility); not significant in 35 days
- Gold: no test rejects at 5%; mean of x_t is -0.34 (missed downward drift); one 1% hit, -3.78% on Monday 28 September

The Portfolio: Correlation Is the Unstable Input

60/40 S&P 500/TLT plus Bitcoin Bitcoin weight	volatility (%, ann.) predicted	 realised	Bitcoin share of risk (%) predicted	 realised
0%	12.2	8.1	–	–
5%	12.8	8.7	11.9	13.2
10%	14.1	9.6	29.4	26.6

- Daily simple returns; TLT adjusted close, S&P 500 price index
- Correlation Bitcoin – S&P 500: 0.29 in 2018–2026, 0.76 over the 25 new trading days
- Bitcoin's own volatility halved (63.3% to 31.4% a year), yet its share of risk changed little: the higher correlation offset the lower volatility

$$\text{share}_{\mathbf{BTC}} = \frac{w_B\,(\Sigma w)_B}{w'\Sigma w}, \qquad (\Sigma w)_B = w_B\sigma_B^2 + \textstyle\sum_{j\neq B} w_j\rho_{Bj}\sigma_B\sigma_j$$

- The risk budget is a forecast of the correlation, not only of the volatilities